Two-Digit Multiplication with Area Models
Answer Key

Grade 4–5 — Number and Operations in Base Ten

Quick Answers

1. 322, 322
2. 480, 480
3. 884, 884
4. 1600
5. 1081, 1081
6. 24, 1404, 1404
7. 1020, 1020
8. 1800
9. 1665, 1665
10. 864, 864

Curriculum

Level: Grade 4–5 — Number and Operations in Base Ten
4.NBT.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 19 tagged checks.

Common wrong answers

4. *If they got 1680: rounded only the first factor and multiplied by the exact second factor*
6. *If they got 1380: added only the three partial products already printed and never filled the empty region*
8. *If they got 1200: rounded 29 down to 20 instead of up to 30*
10. *If they got 624: multiplied tens by tens and ones by ones only, skipping the two cross regions*

How to grade this. The four partial products are the work; the sum is only the last line. If a student's total is wrong, look first at whether all four regions were filled — the classic error is keeping only tens × tens and ones × ones.

1. 23×14, model split as $20 + 3$ across and $10 + 4$ down.
Solution: Fill the four regions, then add.

$$20 \times 10 = 200 \qquad 3 \times 10 = 30 \qquad 20 \times 4 = 80 \qquad 3 \times 4 = 12$$

Adding the partial products: $200 + 30 + 80 + 12 = 322$, and the model is the whole rectangle, so $23 \times 14 = \boxed{322}$.

2. 32×15, model split as $30 + 2$ across and $10 + 5$ down.
Solution:

$$30 \times 10 = 300 \qquad 2 \times 10 = 20 \qquad 30 \times 5 = 150 \qquad 2 \times 5 = 10$$

$300 + 20 + 150 + 10 = 480$, so $32 \times 15 = \boxed{480}$.

3. 26×34, model split as $20 + 6$ across and $30 + 4$ down.
Solution: Both cross regions are large here, which is exactly why they cannot be skipped.

$$20 \times 30 = 600 \qquad 6 \times 30 = 180 \qquad 20 \times 4 = 80 \qquad 6 \times 4 = 24$$

$600 + 180 + 80 + 24 = 884$, so $26 \times 34 = \boxed{884}$.

4. Estimate 38×42 by rounding each factor to the nearest ten.
Solution: 38 is nearer to 40 than to 30; 42 is nearer to 40 than to 50. Rounding *both* factors gives 40×40, and $4 \times 4 = 16$ tells you the estimate is 16 hundreds: $\boxed{1600}$. (The exact product is 1596, so the estimate is a good size check.)

5. 47×23, model split as $40 + 7$ across and $20 + 3$ down.
Solution:

$$40 \times 20 = 800 \qquad 7 \times 20 = 140 \qquad 40 \times 3 = 120 \qquad 7 \times 3 = 21$$

$800 + 140 + 120 + 21 = 1081$, so $47 \times 23 = \boxed{1081}$.

6. 54×26 with three regions printed (1000, 80, 300) and one region empty.
Solution: The empty region is the row labelled 6 meeting the column labelled 4, so it holds $4 \times 6 = \boxed{24}$.
Now add all four regions: $1000 + 300 + 80 + 24 = 1404$, so $54 \times 26 = \boxed{1404}$.

7. 68×15, model split as $60 + 8$ across and $10 + 5$ down.
Solution:

$$60 \times 10 = 600 \qquad 8 \times 10 = 80 \qquad 60 \times 5 = 300 \qquad 8 \times 5 = 40$$

$600 + 80 + 300 + 40 = 1020$, so $68 \times 15 = \boxed{1020}$.

8. Estimate 63×29 by rounding each factor to the nearest ten.
Solution: 63 rounds down to 60. The second factor is the trap: 29 is one away from 30 and nine away from 20, so it rounds *up*. The estimate is $60 \times 30 = \boxed{1800}$. (Exact: 1827.)

9. 45×37, student supplies the split.
Solution: Split each factor by place value: $45 = 40 + 5$ and $37 = 30 + 7$. That labels the model 40 and 5 across, 30 and 7 down.

$$40 \times 30 = 1200 \qquad 5 \times 30 = 150 \qquad 40 \times 7 = 280 \qquad 5 \times 7 = 35$$

$1200 + 280 + 150 + 35 = 1665$, so $45 \times 37 = \boxed{1665}$.

10. Maya's model for 24×36: she wrote only tens $\times$ tens and ones $\times$ ones and answered 624.

Solution: Maya found $20 \times 30 = 600$ and $4 \times 6 = 24$, then stopped. She left out the two cross regions, the ones $\times$ tens and tens $\times$ ones rectangles:

$$4 \times 30 = 120 \qquad 20 \times 6 = 120$$

All four regions together give $600 + 120 + 120 + 24 = 864$, so $24 \times 36 = \boxed{864}$. A one-sentence answer: *she multiplied only the matching parts and skipped the two rectangles where tens meet ones.* The estimate $20 \times 40 = 800$ also shows 624 is far too small.